

Post-hoc Uncertainty Aware Refinement for Monocular Depth Estimation

Zhiang Chen

Julian Elyes

Longxiang Jiao

Kai Wen Li

Department of Computer Science, ETH Zurich

Abstract

Foundation models have been shown to be extremely powerful in the context of monocular depth estimation. Despite this, SOTA models can exhibit erroneous behavior on occlusions and featureless areas simply due to the problems ill-posed nature. As such, we target the regions of uncertainty in the prediction, which often corresponds to a majority of the prediction error. We propose a deep ensemble method to estimate the uncertainty in foundation models post-hoc and augment an adaptive depth completion framework to refine the base prediction using uncertainty.

1. Introduction

Monocular depth estimation (MDE) aims to predict the distance of a pixel to the camera using a single image. This problem is considered to be highly ill-posed due to the lack of reference observation as in the stereo vision and high sensitivity to lighting. Yet with the emergence of neural networks this has been largely overcome using CNNs [15] and recently, transformer-based architectures [16].

In this work we build on top of existing architectures to refine their predictions. This amounts to identifying areas of failure, then extracting local and historical patterns to make new predictions. As seen in Fig. 1, these areas of failure can be characterized as areas of uncertainty, where more variance can be found in the models predictions. As such, we take an interest in modeling the pixels of a depth estimate as univariate Gaussian distributions.

Furthermore, we take inspiration from depth completion which refines noisy predictions based on sparse measurements. In our setting, the pixels of high certainty can be seen as accurate measurements, with a continuous variance map instead of a binary mask. We also adopt the notion of iteratively propagating refinements using affinity learning. In particular we contribute: 1) A deep ensemble approach to quantify uncertainty post-hoc, 2) An uncertainty aware depth refinement framework.

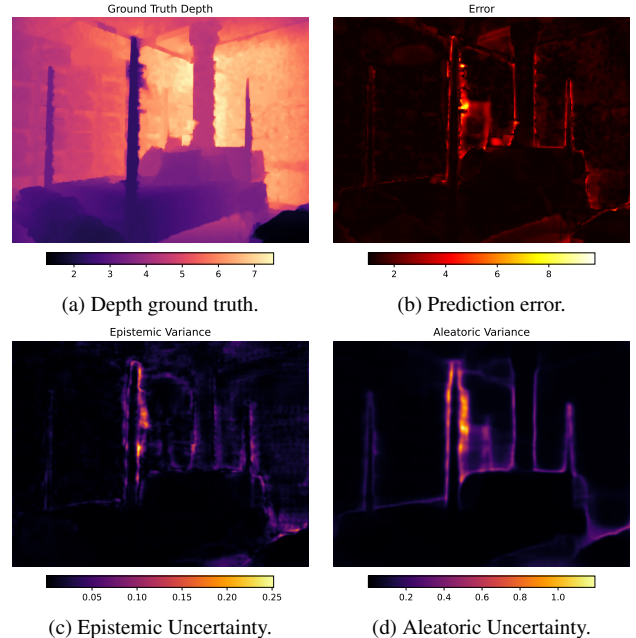


Figure 1. Correspondence of Uncertainty to Error

1.1. Foundation Models

The zero-/few-shot learning capabilities of foundation models relies on large-scale datasets and effective training paradigms. MiDaS [3, 15] introduces dataset mixing and the scale-shift invariant loss. ZoeDepth [2] uses domain specific adaptive binning to enable joint training. DepthAnything [18, 19] captures generalization using unlabeled data and knowledge distillation. Other approaches include Marigold [7] and Unidepth [13] which leverages latent diffusion and camera representation respectively.

1.2. Uncertainty Quantification

Model uncertainty can be decomposed into *epistemic* and *aleatoric* uncertainty. Epistemic uncertainty is reducible, largely arising from uncertainty in the model and decreases with more data. Whereas aleatoric uncertainty is irreducible and corresponds to the inherent noise in the measurements. Many methods exist to model epistemic uncertainty such

as Monte Carlo dropout and Bayesian Neural Networks [1, 8]. Unlike deep ensembles [9], these methods either elicit unreliable estimates or are computationally infeasible. In MDE there are also other variants such as image flipping, self distillation [14] and gradient analysis [5]. For a full treatment of methods and benchmarks we refer to [14].

1.3. Affinity Learning and Depth Refinement

Affinity propagation is a clustering technique involving quantifying similarities between data points then propagating the qualities of the most suitable exemplars. In depth refinement, this constitutes learning a similarity function between a pixel and its neighbors then propagating their values as a weighted sum. These spatial propagation networks adaptively induce diffusive sharpening and can be convolution-[4, 11, 12] or attention-based [6, 20].

2. Methodology

2.1. Post-hoc Uncertainty Modelling

Given a pretrained, general purpose architecture ψ generating a depth map $\mathbf{D} \in \mathbb{R}^{H \times W}$ from an RGB image $\mathbf{I} \in \mathbb{R}^{3 \times H \times W}$, we estimate the uncertainty of its prediction as heteroscedastic variance. We follow the procedure described in [8, 9, 14] and finetune a deep ensemble of foundation models to output a pixelwise distribution $p(y|x, \mathcal{D})$ for $\mathcal{D} = \{\mathbf{I}_i, \mathbf{D}_i\}_{i=1}^N$. Instead of co-optimizing for prediction and uncertainty quantification, the noise of each prediction $\mathbf{D} = \psi(\mathbf{I}) + \mathcal{N}(0, \sigma^2)$ is learned after pretraining (i.e. post-hoc) to remove the predictive dependence on uncertainty and to retain performance.

We first finetune the entire encoder-decoder architecture with the shift invariant log loss (SILL) then duplicate the decoder to train for the Gaussian negative log-likelihood (NLL). Lastly, a variance decomposition is obtained by averaging over the ensemble as a Gaussian mixture $p(y|x, \mathcal{D}) = M^{-1} \sum_{m=1}^M p_m(y|x, \psi_m, \mathcal{D})$ and collecting the epistemic and aleatoric uncertainty terms

$$\mu(x) = M^{-1} \sum_{m=1}^M \mu_m(x) \quad (1)$$

$$\begin{aligned} \sigma^2(x) &= M^{-1} \sum_{m=1}^M (\sigma_m^2(x) + \mu_m^2(x)) - \mu^2(x) \\ &\approx \underbrace{\mathbb{E}_{\psi_m|\mathcal{D}}[\sigma_m^2(x)]}_{\text{aleatoric}} + \underbrace{\text{Var}_{\psi_m|\mathcal{D}}[\mu_m(x)]}_{\text{epistemic}} \end{aligned} \quad (2)$$

where $\sigma_m^2(x)$ and $\mu_m(x)$ are the variance and mean of model ψ_m at location x for $p(y|x, \mathcal{D}) \sim \mathcal{N}(\mu(x), \sigma^2(x))$. Note that in practice instead of NLL we train for the $\beta - NLL$ objective [17]

$$\mathcal{L}_{\beta-NLL} = (\bar{\sigma}^2)_{\text{nograd}}^\beta \cdot \left(\frac{\log \sigma^2}{2} + \frac{(x - \mu(x))^2}{2\sigma^2} \right) \quad (3)$$

which is said to counteract the imbalance between optimizing high and low uncertainty predictions.

2.2. Depth Refinement

Our depth refinement pipeline rests heavily on the field of depth completion, where given sparse ground truth measurements $\mathbf{S} \in \mathbb{R}^{H \times W}$, the remaining predictions are iteratively improved. In our case we treat the low uncertainty predictions as measurements via thresholding and implement an augmented version of the sparse depth refinement model (SDR) from [6]. The pipeline consists of:

1. The foundation ensemble and variance decoder outputting the prediction $\mathbf{D}^0$ and uncertainty map $\mathbf{P}^0$

$$\mathbf{D}^0 = [\mu(x)]_{x \in I} \quad \mathbf{P}^0 = [\sigma^2(x)]_{x \in I}. \quad (4)$$

2. A guidance network Θ generating features

$$\mathbf{G} = \Theta(\mathbf{I}, \mathbf{S}, \mathbf{D}^0) \quad (5)$$

$$\mathbf{S} = [\mathcal{N}(\mu(x), \sigma^2(x)) \cdot \mathbb{I}\{\sigma^2(x) \leq f_{\sigma^2}(\tau)\}]_{x \in I} \quad (6)$$

where $\mathbf{S}$ is the sparse "measurement" from sampling at low uncertainty regions with tolerance τ using $f_{\sigma^2}(\tau) = \tau \max_x \sigma^2(x) + (1 - \tau) \min_x \sigma^2(x)$ to be robust against different variance magnitudes.

3. The masked spatial propagation network (MSPN)

$$\Phi(\mathbf{D}^n, \mathbf{P}^n, \mathbf{G}) = (\hat{\mathbf{D}}^{n+1}, \hat{\mathbf{P}}^{n+1}). \quad (7)$$

We use finetuned DepthAnythingV2 [19] models as our foundation ensemble. The pretrained guidance network from SDR is composed of a Pyramid ViT encoder and a convolution decoder with NAF blocks for feature extraction. The MSPN, also from SDR, is finetuned and consists of spatial attention blocks which maps the initial input to key and query features

$$\mathbf{K}^n = f_K([\mathbf{D}^n, \mathbf{G}]), \mathbf{Q}^n = f_Q([\mathbf{D}^n, \mathbf{G}]) \odot \frac{1}{1 + \mathbf{P}^n} \quad (8)$$

before computing per pixel neighborhood attention scores

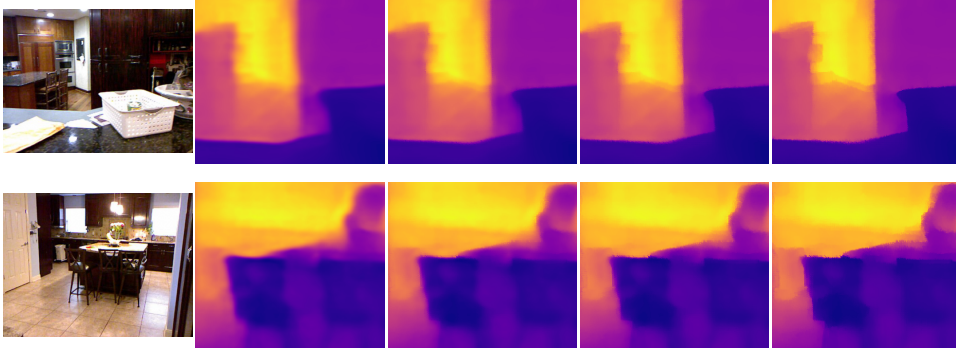
$$a_{i,j} = \text{softmax}((\mathbf{Q}_{i,j}^n)^T N_p^{i,j}(\mathbf{K}^n) + b) \in \mathbb{R}^{p^2} \quad (9)$$

where $N_p^{i,j}(\cdot)$ extracts the neighboring pixels in a window of size $p \times p$ centered at i, j . As in typical affinity learning settings with spatial propagation networks [4, 20], the attention scores act as the filter weights of a convolution operation with a window of the input map

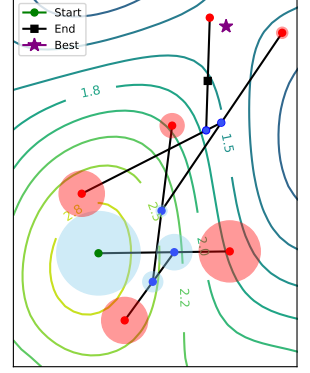
$$\hat{\mathbf{D}}_{i,j}^{n+1} = \sum_{k=1}^{p^2} a_{i,j,k} \cdot N_p^{i,j}(\mathbf{D}^n)_k. \quad (10)$$

We consider this refinement as the mean of a Gaussian mixture with mixture weights $a_{i,j,k}$

$$\begin{aligned} p_{i,j}^{n+1}(y|x, \mathcal{D}) &= \sum_{k=1}^{p^2} a_{i,j,k} \cdot p_{i(k)}^n(y|x, \mathcal{D}) \\ &\sim \mathcal{N}(\hat{\mathbf{D}}_{i,j}^{n+1}, \hat{\mathbf{P}}_{i,j}^{n+1}) \end{aligned} \quad (11)$$



(a) Uncertainty-aware depth refinement has a sharpening effect.



(b) Potential UDR convergence.

where $\iota(k)$ indexes the k -th element in the window centered at (i, j) . Lastly, given the neighbor pixel distributions $p_{\iota(k)}^n(y|\mathbf{x}, \mathcal{D}) \sim \mathcal{N}(N_p^{i,j}(\mathbf{D}^n)_k, N_p^{i,j}(\mathbf{P}^n)_k)$ we provide the MSPN uncertainty estimate as the variance of the resulting Gaussian mixture

$$\hat{\mathbf{P}}_{i,j}^{n+1} = \sum_{k=1}^{p^2} a_{i,j,k} \cdot N_p^{i,j}(\mathbf{P}^n + (\mathbf{D}^n)^2)_k - (\hat{\mathbf{D}}_{i,j}^n)^2. \quad (12)$$

2.3. Uncertainty Propagation

The key modification of our work is to extend the discrete binary masking from depth completion to a continuous setting, where our uncertainty map comprises per pixel variances. We adopt the uncertainty aware perspective of Kalman filters [10], which predicts the state of a system by linearly interpolating between the estimate and some measurement based on the Kalman gain κ in each iteration. This exhibits variance optimality under the assumption of linear process and measurement models, which is discussed further in Sec. 3.4.

In our case, we model depth estimation as a static process by viewing each pixel as independent one-dimensional states and the MSPN refinement as a noisy measurement model dependent only on the previous depth estimate. We retrieve furthermore the Kalman update scheme:

$$\kappa^n = \mathbf{P}^n / (\mathbf{P}^n + \hat{\mathbf{P}}^{n+1}) \quad (13)$$

$$\mathbf{D}^{n+1} = \mathbf{D}^n + \kappa^n \odot (\hat{\mathbf{D}}^{n+1} - \mathbf{D}^n) \quad (14)$$

$$\mathbf{P}^{n+1} = (1 - \kappa^n) \odot (\mathbf{P}^n + \varepsilon) \quad (15)$$

where $\odot, /$ are coefficient-wise multiplication and division, and ε is some Gaussian process noise. To ensure that the MSPN is optimized with respect to this propagation scheme, we train it on the SILL objective with backpropagation through time. The potential convergence behavior in two-dimensions is illustrated in Fig. 2b. Starting from an **estimate**, the **refinement** dictates the direction of improvement and the uncertainty (circle) determines the step size.

3. Experimental Results

3.1. Uncertainty to Error Correspondence

We evaluate the uncertainty calibration performance of the ensemble with *Area Under Sparsification Error* (AUSE) and *Area Under Random Gain* (AURG) as seen in Tab. 1. Given a metric ϵ , we construct sparsification plots by first sorting the pixel wise prediction error (oracle) in a depth map $\mathbf{D}$ with descending order. Then we sort the predictions in descending uncertainty (estimate) and iteratively remove samples whilst evaluating ϵ to build the plots. Thus, if the oracle errors correspond to high uncertainty pixels then the difference between the two plots should be small (AUSE), and similarly the difference from the estimate plot to a random sparsification (AURG) should be large.

Method	AUSE $_{L_1}^\downarrow$	AUSE $_{L_2}^\downarrow$	AURG $_{L_1}^\uparrow$	AURG $_{L_2}^\uparrow$
Flipping	0.0545	0.0265	0.0316	0.0302
Ours	0.0297	0.0082	0.0563	0.0486

Table 1. Quantitative uncertainty estimation, $\epsilon \in \{L_1, L_2\}$

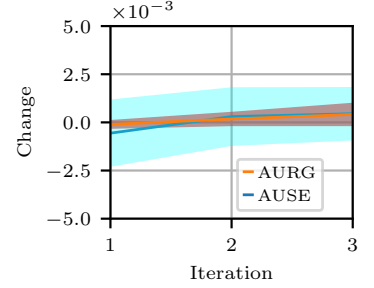
We compare our method against a simple yet well known approach, image flipping. By horizontally flipping the image $\mathbf{I}_T = T(\mathbf{I})$ before prediction then reversing, the uncertainty can be quantified by the error $|\psi(\mathbf{I}) - T^{-1}(\psi(\mathbf{I}_T))|$.

3.2. Depth Refinement

In Fig. 3a we assess the performance of base foundational models, base SDR with confidence thresholded binary masks, finetuned ensemble and the uncertainty-aware depth refinement model (UDR) on MDE using metrics summarized in Tab. 2 where the image has size $N = HW$, d_i, d_i^* are the i -th prediction or ground truth respectively and $\alpha = \sum_{i=1}^N d_i d_i^* / \sum_{i=1}^N d_i^2$. Additionally, the qualitative effect of spatial propagation networks is displayed in Fig. 2a, where blurry, high variance predictions are sharpened by learned filters.

Method	si-RMSE↓	AbsRel↓	RMSE↓	logRMSE↓	$\delta 1 \uparrow$	$\delta 2 \uparrow$	$\delta 3 \uparrow$
BaseDA2[19]	0.565	0.357	1.076	0.375	0.416	0.728	0.896
MariGold[7]	0.407	0.171	0.504	0.253	0.738	0.917	0.962
MiDaS[3]	0.580	0.360	1.044	0.422	0.469	0.754	0.890
SDR[6]	0.2718	0.0556	0.2837	0.0865	0.9653	0.9932	0.9985
EnsembleDA2	0.2069	0.0487	0.2196	0.0698	0.9814	0.9973	0.9995
UDR (ours)	0.2068	0.0479	0.2197	0.0696	0.9816	0.9972	0.9994

(a) Comparison of MDE results: UDR surpasses the ensemble (barely) whereas SDR performs a lot worse.



(b) Average change in uncertainty

3.3. Ablation

Kalman gain evolution: We verify that the Kalman filter is valid for combining uncertainties. In a typical use case, the Kalman gain starts at 1, favoring measurements, then drops drastically as the prediction uncertainty decreases. In Fig. 4 we visualize the evolution of the per pixel clipped inverse logarithmic gain $\kappa_{\text{cil}} = \min(-\log \kappa, 1)$, which takes values close to 1 if the current prediction is certain and 0 otherwise. Clearly, most of the current predictions are confident, with Kalman gain decrease restricted to occlusion regions such as on the edge of objects. This implies additionally that the total gain likewise decreases.

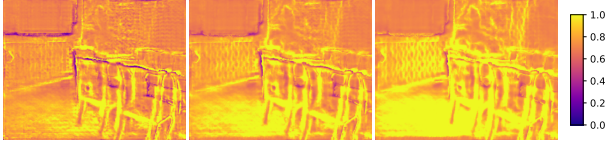


Figure 4. Evolution of κ_{cil} in uncertain regions.

Variance Estimate: While in Sec. 3.1 we showed that the initial calibration performs well, inaccuracy of subsequent iterations can lead to large variance discrepancies. If the AUSE, AURG scores with $\epsilon = L_2$ decreases or increases respectively or if the difference $\delta_{\text{SE}}, \delta_{\text{RG}}$ across iterations remains minimal, then the estimate should be at least as good as the initial estimate. Fig. 3b plots the mean $\delta_{\text{SE}}, \delta_{\text{RG}}$ evolution over 32 images with $\pm\sigma$. Both values are zero in expectation whereas δ_{SE} is more noisy with a slight positive trajectory. Although not ideal, this supports our understanding that inaccurate variance estimates are far from being random yet might not reflect the error well.

si-RMSE	$N^{-1} \sum_{i=1}^N (\alpha d_i - d_i^*)^2$,
AbsRel	$N^{-1} \sum_{i=1}^N d_i / d_i - d_i^* $
logRMSE	$[N^{-1} \sum_{i=1}^N (\log d_i - \log d_i^*)^2]^{1/2}$
δ_k	% of d_i , $\max(d_i/d_i^*, d_i^*/d_i) < 1.25^k$

Table 2. Summary of depth estimation metrics.

3.4. Discussion

The performance of our method relies on four main assumptions: (1) the post-hoc uncertainty estimates corresponds to the error, (2) the predictions are correctly updated based on the combination of uncertainties, (3) the uncertainty of each refinement is accurately estimated, and (4) the MSPN model is capable of improving prediction performance.

From the results in Sec. 3.1 we see that (1) holds. For (2), the linear requirement for the global optimality of Kalman Filters is violated by the variance update Eq. (12). Other extensions such as the Extended or Unscented Kalman Filter [10] can achieve local optimality for nonlinear systems with Gaussian guarantees. These methods are nevertheless out of scope and we instead opt to empirically demonstrate the viability of our approach as seen in Sec. 3.3.

The last two points proved to be the most difficult. (3) requires approximating the variance through a spatial attention network. The use of Gaussian mixtures with affinity propagation is thematic to clustering and seems to be supported by experiments in Sec. 3.3 and comparisons of various analytical expressions under simplifying assumptions.

Finally, we were not able to achieve satisfactory results on (4). Due to the extremely large computation graph of the MSPN, model training could not be resolved on the entire dataset with sufficient epochs and iterations. The high performance of the ensemble also hinders the refinement capabilities (lack of improvement possibility), and suggests that there might be some data leakage in the custom dataset.

3.5. Conclusion

We presented a novel extension to depth refinement leveraging the uncertainty in the predictions of foundation models. In our framework, ensemble predictions and uncertainty estimates are refined iteratively using Kalman filters and spatial propagation networks. Although not all results were ideal, secondary experiments showed that our method makes accurate variance estimates based on Gaussian mixture assumptions and considers uncertainty minimizing refinements. A future work could entail knowledge distillation of the MSPN to alleviate the computational load.

References

- [1] Moloud Abdar, Farhad Pourpanah, Sadiq Hussain, Dana Rezazadegan, Li Liu, Mohammad Ghavamzadeh, Paul Fieguth, Xiaochun Cao, Abbas Khosravi, U Rajendra Acharya, et al. A review of uncertainty quantification in deep learning: Techniques, applications and challenges. *Information fusion*, 76:243–297, 2021. [2](#)
- [2] Shariq Farooq Bhat, Reiner Birkel, Diana Wofk, Peter Wonka, and Matthias Müller. Zoedepth: Zero-shot transfer by combining relative and metric depth. *arXiv preprint arXiv:2302.12288*, 2023. [1](#)
- [3] Reiner Birkel, Diana Wofk, and Matthias Müller. Midas v3.1—a model zoo for robust monocular relative depth estimation. *arXiv preprint arXiv:2307.14460*, 2023. [1](#), [4](#)
- [4] Xinjing Cheng, Peng Wang, and Ruigang Yang. Depth estimation via affinity learned with convolutional spatial propagation network. In *Proceedings of the European conference on computer vision (ECCV)*, pages 103–119, 2018. [2](#)
- [5] Julia Hornauer and Vasileios Belagiannis. Gradient-based uncertainty for monocular depth estimation. In *European Conference on Computer Vision*, pages 613–630. Springer, 2022. [2](#)
- [6] Jinyoung Jun, Jae-Han Lee, and Chang-Su Kim. Masked spatial propagation network for sparsity-adaptive depth refinement. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 19768–19778, 2024. [2](#), [4](#)
- [7] Bingxin Ke, Anton Obukhov, Shengyu Huang, Nando Metzger, Rodrigo Caye Daudt, and Konrad Schindler. Repurposing diffusion-based image generators for monocular depth estimation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 9492–9502, 2024. [1](#), [4](#)
- [8] Alex Kendall and Yarin Gal. What uncertainties do we need in bayesian deep learning for computer vision? *Advances in neural information processing systems*, 30, 2017. [2](#)
- [9] Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. *Advances in neural information processing systems*, 30, 2017. [2](#)
- [10] Qiang Li, Ranyang Li, Kaifan Ji, and Wei Dai. Kalman filter and its application. In *2015 8th international conference on intelligent networks and intelligent systems (ICINIS)*, pages 74–77. IEEE, 2015. [3](#), [4](#)
- [11] Sifei Liu, Shalini De Mello, Jinwei Gu, Guangyu Zhong, Ming-Hsuan Yang, and Jan Kautz. Learning affinity via spatial propagation networks. *Advances in Neural Information Processing Systems*, 30, 2017. [2](#)
- [12] Jinsun Park, Kyungdon Joo, Zhe Hu, Chi-Kuei Liu, and In So Kweon. Non-local spatial propagation network for depth completion. In *Computer Vision—ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part XIII 16*, pages 120–136. Springer, 2020. [2](#)
- [13] Luigi Piccinelli, Yung-Hsu Yang, Christos Sakaridis, Mattia Segu, Siyuan Li, Luc Van Gool, and Fisher Yu. Unidepth: Universal monocular metric depth estimation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 10106–10116, 2024. [1](#)
- [14] Matteo Poggi, Filippo Aleotti, Fabio Tosi, and Stefano Mattoccia. On the uncertainty of self-supervised monocular depth estimation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 3227–3237, 2020. [2](#)
- [15] René Ranftl, Katrin Lasinger, David Hafner, Konrad Schindler, and Vladlen Koltun. Towards robust monocular depth estimation: Mixing datasets for zero-shot cross-dataset transfer. *IEEE transactions on pattern analysis and machine intelligence*, 44(3):1623–1637, 2020. [1](#)
- [16] René Ranftl, Alexey Bochkovskiy, and Vladlen Koltun. Vision transformers for dense prediction. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 12179–12188, 2021. [1](#)
- [17] Maximilian Seitzer, Arash Tavakoli, Dimitrije Antic, and Georg Martius. On the pitfalls of heteroscedastic uncertainty estimation with probabilistic neural networks. *arXiv preprint arXiv:2203.09168*, 2022. [2](#)
- [18] Lihe Yang, Bingyi Kang, Zilong Huang, Xiaogang Xu, Jiashi Feng, and Hengshuang Zhao. Depth anything: Unleashing the power of large-scale unlabeled data. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 10371–10381, 2024. [1](#)
- [19] Lihe Yang, Bingyi Kang, Zilong Huang, Zhen Zhao, Xiaogang Xu, Jiashi Feng, and Hengshuang Zhao. Depth anything v2. *Advances in Neural Information Processing Systems*, 37:21875–21911, 2024. [1](#), [2](#), [4](#)
- [20] Youmin Zhang, Xianda Guo, Matteo Poggi, Zheng Zhu, Guan Huang, and Stefano Mattoccia. Completionformer: Depth completion with convolutions and vision transformers. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 18527–18536, 2023. [2](#)

Post-hoc Uncertainty Aware Refinement for Monocular Depth Estimation

Supplementary Material

4. Appendix

4.1. Derivations

4.1.1. Uncertainty Decomposition

We utilized the law of total variance to derive the aleatoric and epistemic uncertainty, from which the formula for the mixture variance naturally followed.

$$\begin{aligned}\text{Var}[X] &= \mathbb{E}[\text{Var}(X|Z)] + \text{Var}[\mathbb{E}(X|Z)] \\ &= \mathbb{E}[\sigma^2(Z)] + \text{Var}[\mu(Z)] \\ &= \sum_k \pi_k \sigma_k^2 + \sum_k \pi_k (\mu_k - \bar{\mu})^2 \\ &= \sum_k \pi_k (\sigma_k^2 + \mu_k^2) - \bar{\mu}^2\end{aligned}\quad (16)$$

where $P(Z = k) = \pi_k$, $X|Z \sim \mathcal{N}(\mu(Z), \sigma^2(Z))$ and $X|Z = k \sim \mathcal{N}(\mu_k, \sigma_k^2)$. In the special case where Z is uniformly distributed, we have Eq. (2).

4.1.2. Variance Propagation Analysis

The rationale behind choosing the Gaussian mixture was to decouple the variance estimation process from the model training of MSPN. We attempted first to train a small variance network alongside the MSPN, which resulted in relatively accurate variance predictions but severely limited the refinement capability and complicated training.

As mentioned in Sec. 3.4, we aim to discover a similar analytical form for the variance estimate of the refinement Eq. (10) by making simplifying assumptions. Clearly the estimate takes the form $\sum_{k=1}^{p^2} \Phi_k(a_{i,j}, \mathbf{D}^n, \mathbf{P}^n)$, and we look for alternative functions $\Phi_k(a, x, p) = \phi(a_k, N_p^{i,j}(x), N_p^{i,j}(p))$ including:

1. The base MSPN estimate $\phi_{\text{SDR}}(a, x, p) = ax$
2. Treating the refinement as a linear combination of random variables $\phi_{\text{lin}}(a, x, p) = a^2x$
3. Empirically determined $\phi_{\text{root}}(a, x, p) = a^{1/2}x$
4. Modeling the attention $a_{i,j} = f(\mathbf{D}_{i,j}^n)$ with
 - (a) assuming linear $f : x \mapsto ax + b$
 - (b) assuming f non-linear, differentiable

(1) underestimates the variance as seen in Fig. 3a, (2) even more so with a gain ≈ 0 in each iteration. While in (3), the min and max gain converged, the mean did not. All these methods failed to consider the previous variance, which seems to be vital. For (4), we derived the bounds in (a)

$$\begin{aligned}\text{Var}[f(x)x] &= a^2\text{Var}[x^2] + b\sigma^2 \\ &\leq a^2(2\sigma^2(\sigma^2 + 2\mu^2)) + b\sigma^2\end{aligned}\quad (17)$$

using that $\mathbb{E}[x^4] = \mu^4 + 6\mu^2\sigma^2 + 3\sigma^4$ and $\mathbb{E}[x^2] = \mu^2 + \sigma^2$.

For (b) we noted that $\forall x. f(x) \in [0, 1]$ in order to obtain

$$\begin{aligned}\text{Var}[f(x)x] &\leq \mathbb{E}[x^2] - \mathbb{E}[f(x)x]^2 \\ &= \sigma^2 + \mu^2 - (\mu\mathbb{E}[f(x)] + \sigma^2\mathbb{E}[f'(x)]) \\ &\approx \sigma^2 + \mu^2 - (\mu f(\mu) + C\sigma^2)\end{aligned}\quad (18)$$

where we used Stein's lemma and the 1st. Taylor approximation of $\mathbb{E}[f(x)]$. Although both bounds were infeasible to compute due to the inaccessibility of the gradient $\nabla_{\mathbf{D}} a_{i,j,k}$, we recognized that the term $\sigma^2 + \mu^2$ was required. This however vastly overestimated the variance and required a similarly large penalty to bring it to the appropriate range, leading to the Gaussian mixture proposal $\Phi_k(a, x, p) = a_k(p_k + x_k^2) - \sum_k a_k x_k$ in Eq. (12).

4.2. Limitations

The main disadvantage of our method is that the total time required to train the uncertainty ensemble and the MSPN does not lend itself well to continuous experimentation, which led to potential misgivings in the final results. The MSPN initially required more than 20GBs of VRAM to train for six iteration with a batch size of around 4-8 taking 1.5 hours per epoche. We hypothesize that this was due to the complexity of the computational graph required by spatial attention operations. Although we managed to reduce the load, it is unclear how much this affected the capabilities of the network.

4.3. Future Work

To address this, we conceptualized a self learning paradigm, where a smaller CNN or ViT is used to distill the affinity learning behavior from MSPN before finetuning to optimize performance. This most likely would then require a custom loss function to capture the intricacies inherently present in spatial attention networks. We were also not equipped to extensively study non-linear Kalman filters. The extended Kalman filter required gradients of the MSPN which were difficult to procure (as above), and the unscented Kalman filter required $3 \times H \times W$ forward passes of MSPN per iteration, which was also not computationally possible.



Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

Declaration of originality

The signed declaration of originality is a component of every written paper or thesis authored during the course of studies. In consultation with the supervisor, one of the following three options must be selected:

- ☒ I confirm that I authored the work in question independently and in my own words, i.e. that no one helped me to author it. Suggestions from the supervisor regarding language and content are excepted. I used no generative artificial intelligence technologies¹.
- ☐ I confirm that I authored the work in question independently and in my own words, i.e. that no one helped me to author it. Suggestions from the supervisor regarding language and content are excepted. I used and cited generative artificial intelligence technologies².
- ☐ I confirm that I authored the work in question independently and in my own words, i.e. that no one helped me to author it. Suggestions from the supervisor regarding language and content are excepted. I used generative artificial intelligence technologies³. In consultation with the supervisor, I did not cite them.

Title of paper or thesis:

Post-hoc Uncertainty Aware Refinement for Monocular Depth Estimation

Authored by:

If the work was compiled in a group, the names of all authors are required.

Last name(s):

Elyes
Chen
Jiao
Li

First name(s):

Julian
Zhiang
Longxiang
Kai Wen

With my signature I confirm the following:

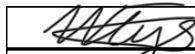
- I have adhered to the rules set out in the Citation Guide.
- I have documented all methods, data and processes truthfully and fully.
- I have mentioned all persons who were significant facilitators of the work.

I am aware that the work may be screened electronically for originality.

Place, date

Zurich, 28/05/2025
Zurich, 24/05/2025
Zurich, 28/05/2025
Zurich, 28/05/2025

Signature(s)






If the work was compiled in a group, the names of all authors are required. Through their signatures they vouch jointly for the entire content of the written work.

¹ E.g. ChatGPT, DALL E 2, Google Bard

² E.g. ChatGPT, DALL E 2, Google Bard

³ E.g. ChatGPT, DALL E 2, Google Bard